**Pseudocode**

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MODULE WelcomeMessage
        DISPLAY “Welcome to JeepTrack!”
END MODULE

MODULE DisplayLocations
        DISPLAY “Available Locations”
        DISPLAY “> San Pablo City                    > SM San Pablo”
        DISPLAY “> LSPU-San Pablo City Campus        > Liliw”
        DISPLAY “>Nagcarlan                          >Tiaong”
        DISPLAY “> Calauan                           >Los Banos”
        DISPLAY “> Sta Cruz”
END MODULE

MODULE PassengerTypes
        DISPLAY “Passenger Type”
        DISPLAY “> Regular                           > Student”
        DISPLAY “> PWD                               >Senior Citizen”
END MODULE

START
        CALL WelcomeMessage

        INITIALIZE locations as ARRAY OF STRING = {“SAN PABLO CITY”, “SM SAN PABLO”, “LSPU-SAN PABLO
        CITY CAMPUS”, “LILIW”,  “NAGCARLAN”, ”TIAONG”, “CALAUAN”, “LOS BANOS”, “STA CRUZ”}
        DECLARE currentLocation AS STRING
        DECLARE destination AS STRING

        WHILE true
                CALL DisplayLocations

                DISPLAY “Enter your current location (enter “exit” to quit): ”
                INPUT currentLocation

                INITIALIZE locationFound = false

                FOR  i = 0 TO LENGTH(locations) - 1
                        IF currentLocation == locations[i] THEN
                                SET locationFound = true
                                EXIT LOOP
                        ELSE IF currentLocation CONVERT TO LOWERCASE == “exit” THEN
                                DISPLAY “Exiting the program…”
                                EXIT PROGRAM
                        END IF
                END FOR

                IF locationFound is true THEN
                        EXIT LOOP
                END IF

                DISPLAY “LOCATION NOT FOUND!”
        END WHILE

        WHILE true
                CALL DisplayLocations

                DISPLAY “Enter your destination (enter “exit” to quit): ”
                INPUT destination
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        INITIALIZE locationFound = false

        IF destination == currentLocation THEN
                DISPLAY “DESTINATION AND CURRENT LOCATION MUST BE DIFFERENT”
                CONTINUE LOOP
        END IF


        FOR  i = 0 TO LENGTH(locations) - 1
                IF destination == locations[i] THEN
                        SET locationFound = true
                        EXIT LOOP
                ELSE IF destination CONVERT TO LOWERCASE == “exit” THEN
                        DISPLAY “Exiting the program…”
                        EXIT PROGRAM

                END IF
        END FOR

        IF locationFound is true THEN
                EXIT LOOP
        END IF

        DISPLAY “LOCATION NOT FOUND!”
END WHILE

INITIALIZE routes = currentLocation + “-” + destination
DECLARE travelDistance AS FLOAT
DECLARE travelTime AS INTEGER

SWITCH routes
        CASE "SAN PABLO CITY-SM SAN PABLO":
        CASE "SM SAN PABLO-SAN PABLO CITY":
                SET travelDistance = 2.9;
                SET travelTime = 9;

        CASE "SAN PABLO CITY-LSPU-SAN PABLO CITY CAMPUS":
        CASE "LSPU-SAN PABLO CITY CAMPUS-SAN PABLO CITY":
                SET travelDistance = 2.9;
                SET travelTime = 11;

        CASE "SAN PABLO CITY-LILIW":
        CASE "LILIW-SAN PABLO CITY":
                SET travelDistance = 19.8;
                SET travelTime = 46;

        CASE "SAN PABLO CITY-NAGCARLAN":
        CASE "NAGCARLAN-SAN PABLO CITY":
                SET travelDistance = 15.8;
                SET travelTime = 37;

        CASE "SAN PABLO CITY-TIAONG":
        CASE "TIAONG-SAN PABLO CITY":
                SET travelDistance = 13.1;
                SET travelTime = 24;

        CASE "LSPU-SAN PABLO CITY CAMPUS-CALAUAN":
        CASE "CALAUAN-LSPU-SAN PABLO CITY CAMPUS":
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                SET travelDistance = 8.3;
                SET travelTime = 15;

        CASE "LSPU-SAN PABLO CITY CAMPUS-LOS BANOS":
        CASE "LOS BANOS-LSPU-SAN PABLO CITY CAMPUS":
                SET travelDistance = 20.7;
                SET travelTime = 40;

        CASE "SAN PABLO CITY-STA CRUZ":
        CASE "STA CRUZ-SAN PABLO CITY":
                SET travelDistance= 30.0;
                SET travelTime = 60;

        CASE "NAGCARLAN-LILIW":
        CASE "LILIW-NAGCARLAN":
                SET travelDistance= 3.2;
                SET travelTime = 9;

        CASE "NAGCARLAN-STA CRUZ":
        CASE "STA CRUZ-NAGCARLAN":
                SET travelDistance = 17;
                SET travelTime = 38;

        DEFAULT:
                DISPLAY “ROUTE NOT FOUND!”
                EXIT PROGRAM
END SWITCH

INITIALIZE passengerTypes AS ARRAY OF STRING = {“REGULAR”, “SENIOR CITIZEN”, “PWD”,
“STUDENT”}
DECLARE passengerType AS STRING

WHILE true
        CALL PassengerTypes

        DISPLAY “Enter passenger type: ”
        INPUT passengerType

        INITIALIZE validPassenger = false

        FOR  i = 0 TO LENGTH(passengerTypes) - 1
                IF passengerType == passengerTypes[i] THEN
                        SET validPassenger = true
                        EXIT LOOP
                END IF
        END FOR

        IF validPassenger is true THEN
                EXIT LOOP
        END IF

        DISPLAY “INVALID PASSENGER TYPE!”
END WHILE

INITIALIZE initialDistance = 4.00
INITIALIZE initialRate = 14.00
INITIALIZE succeedingRate  = 2.00
INITIALIZE discount = 0.20
DECLARE fare AS FLOAT
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DECLARE finalFare AS FLOAT

IF travelDistance <= initialDistance THEN
        SET fare = initialRate
ELSE
        CALCULATE fare = (travelDistance - initialDistance) * succeedingRate + initialRate
END IF

IF passengerType == “REGULAR” THEN
        SET discount = 0
        SET finalFare = fare
ELSE
        CALCULATE finalFare =  fare - ( fare * discount)
END IF

DISPLAY “JEEPTRACK”
DISPLAY “ROUTE”
DISPLAY “DISTANCE                    EST TIME TRAVEL”
DISPLAY “travelDistance km           travelTime minutes”
DISPLAY “PASSENGER TYPE”
DISPLAY “PASSENGER TYPE              DISCOUNT”
DISPLAY “passengerType               discount”
DISPLAY “FARE”
DISPLAY “BOUND TO: currentLocation > destination
DISPLAY “FARE AMOUNT:   ₱ finalFare”
DISPLAY “HAVE A SAFE TRAVEL!”
```